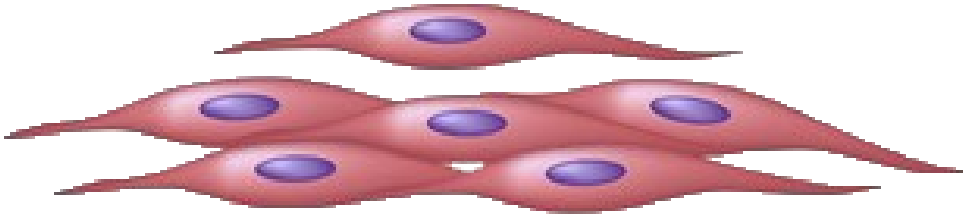
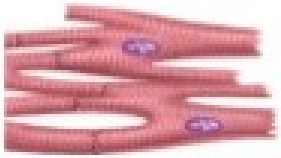
Smooth muscle and Cardiac muscle



Smooth Muscle

Three Types of Muscular Tissue

	Location	Function	Appearance	Control
Skeletal 	skeleton	movement, heat, posture	**striated**, multi-nucleated (eccentric), fibers parallel	**voluntary**
Cardiac 	heart	pump blood continuously	**striated**, one central nucleus	**involuntary**
Visceral (smooth muscle) 	G.I. tract, uterus, eye, blood vessels	Peristalsis, blood pressure, pupil size, erects hairs	**no striations**, one central nucleus	**involuntary**

Cardiac Muscle

- Only in the heart muscle fibers joined together by intercalated discs.
- Network of fibers contracts as a unit self-exciting and rhythmic
- Longer refractory period than skeletal muscle

Structure of cardiac muscle

- Straited muscle fibers
- Sarcolemma present
- Central nucleus
- Contractile proteins like actin, myosin, trponin, tropomyosine all are present.
- Well defined sarcotubular system
- Branched muscle fibers

INTERCALATED DISK

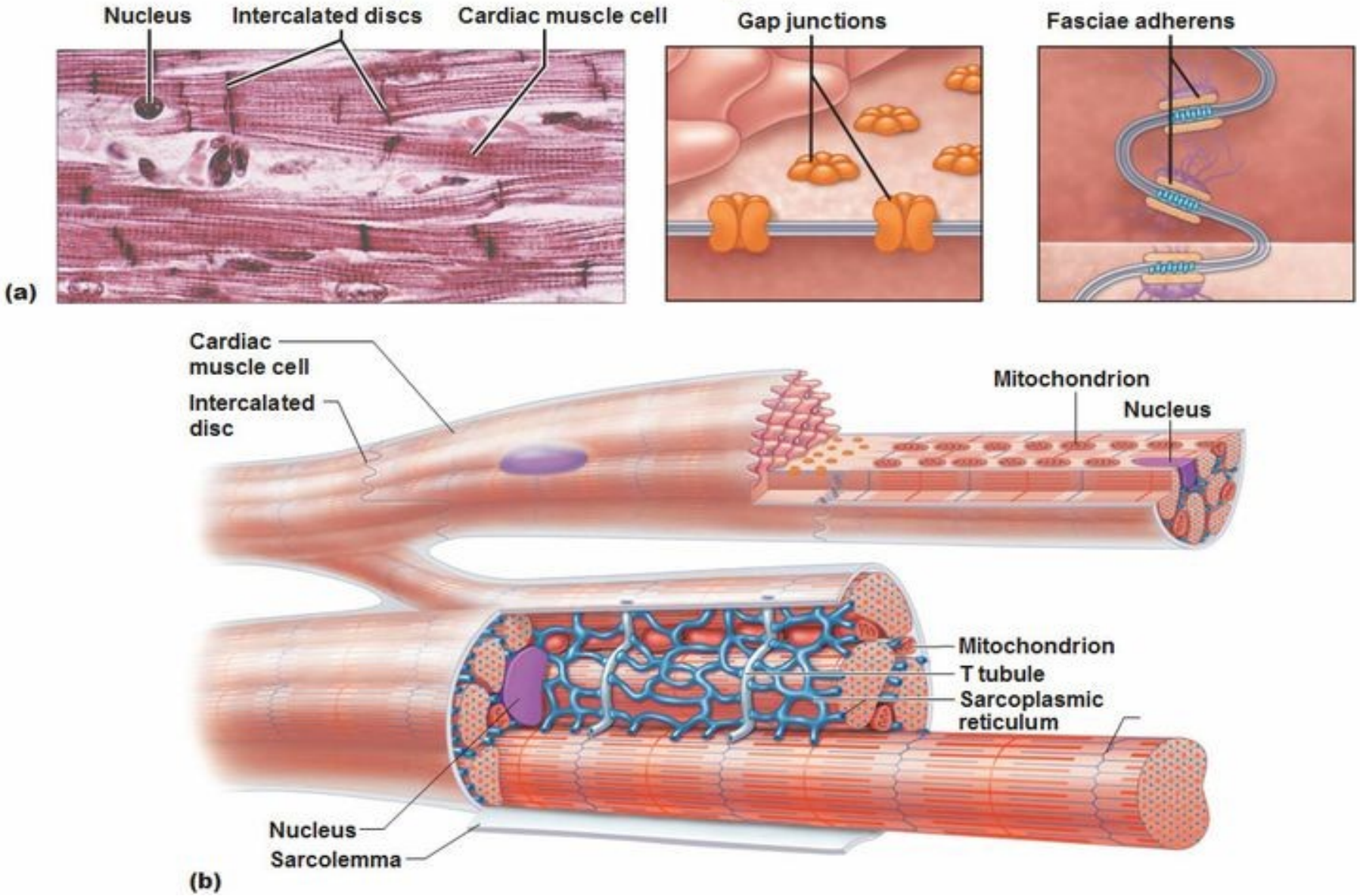
- **At the junctions between branches of neighboring fibers there is a tough intercalated disk.**
- **Disk formed by fusion of membrane of the cardiac muscle branches.**
- **Intercalated disk forms the adherence junctions.**

Syncytium

Cardiac muscle is physiological syncytium, there is no continuity of cytoplasm. But there are gap junction which permit the free flow of ions and facilitate the rapid conduction od action potential.

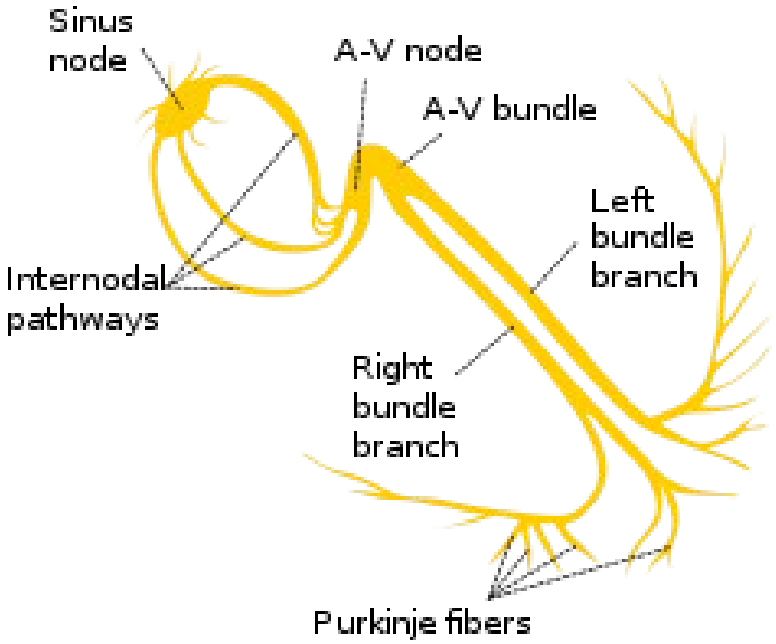
- **Two portions of syncytium**
- **Atrial syncytium**
- **Ventricular syncytium**

Microscopic Anatomy of Cardiac Muscle



CONDUCTING SYSTEM OF HEART

- **Conducting system of heart is formed by modified cardiac muscle fibers.**



SMOOTH MUSCLE

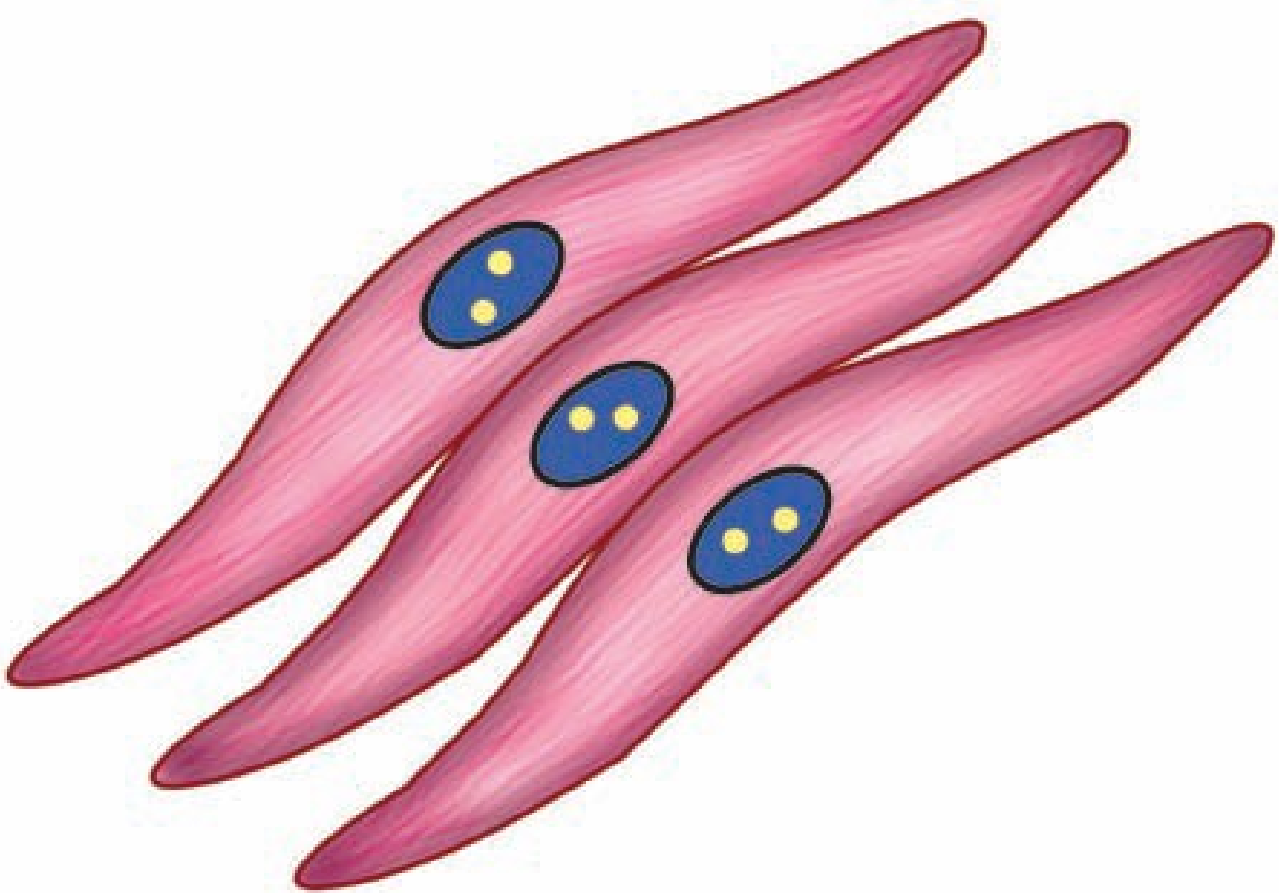
- **Smooth muscles are non-striated and involuntary muscles.**
- **Present in almost all the organs in the form of sheets, bundles or sheaths around other tissues.**
- **Smooth muscles form the major contractile tissues of various organs**

Smooth Muscle Fibers

Compared to skeletal muscle fibers

- **Shorter**
- **Single nucleus**
- **Elongated with tapering ends**
- **Myofilaments randomly organized**
- **No striations**
- **Lack transverse tubules**
- **Sarcoplasmic reticulum not well developed**

STRUCTURE OF SMOOTH MUSCLE



STRUCTURE OF SMOOTH MUSCLE

- Smooth muscle fibers are fusiform or elongated cells.
- These fibers are generally very small, measuring 2 to 5 microns in diameter and 50 to 200 microns in length.
- Nucleus is single and elongated and it is centrally placed.
- Normally, two or more nucleoli are present in the nucleus

STRUCTURE OF SMOOTH MUSCLE

- **Myofibrils and Sarcomere**
- Well-defined myofibrils and sarcomere are absent in smooth muscles.
- So the alternate dark and light bands are absent.
- Absence of dark and light bands gives the non-striated appearance to the smooth muscle.

STRUCTURE OF SMOOTH MUSCLE

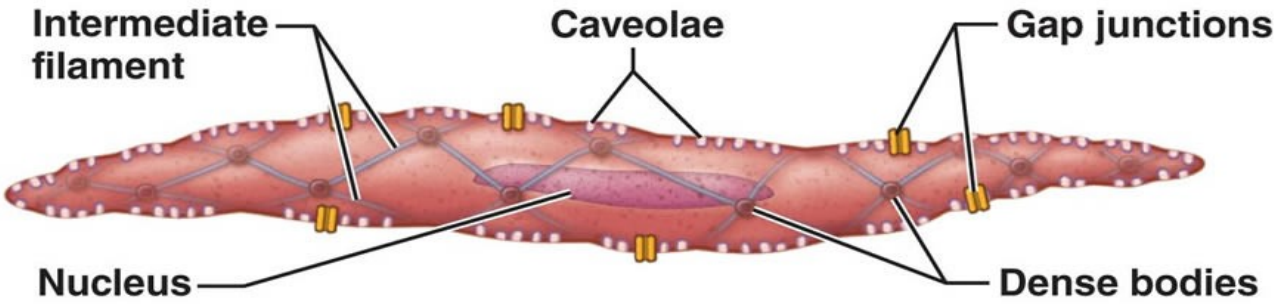
- **Myofilaments and Contractile Proteins**
- Contractile proteins in smooth muscle fiber are actin, myosin and tropomyosin.
- Troponin or troponin-like substance is absent.
- Thick and thin filaments are present in smooth muscle.

STRUCTURE OF SMOOTH MUSCLE

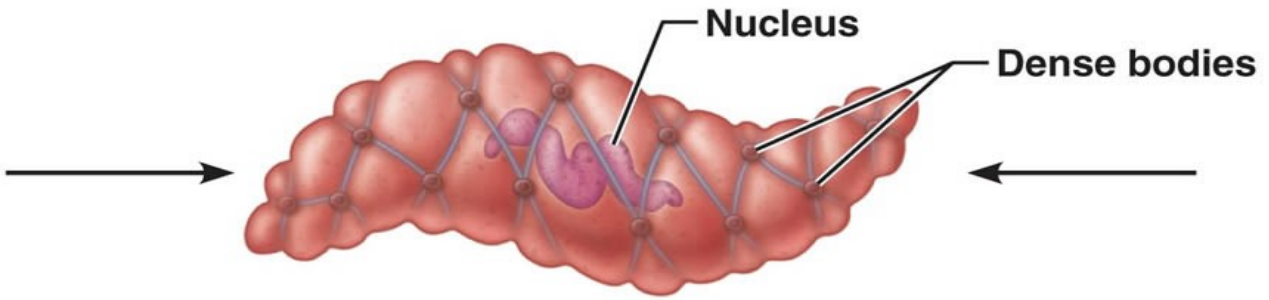
- These filaments are not arranged in
- orderly fashion as in skeletal muscle. Thick filaments are formed by myosin molecules and are scattered in sarcoplasm.
- Thick filaments contain more number
- of cross bridges than in skeletal muscle.
- Thin filaments are formed by actin and tropomyosin molecules.

- ***Dense Bodies***
- Dense bodies are the special structures of smooth muscle fibers to which the actin and tropomyosin molecules of thin filaments are attached.
- Scattered all over the sarcoplasm in the network of intermediate filaments, which is formed by the protein **desmin.**
- Some of the dense bodies are firmly attached with sarcolemma.
- The anchoring of the dense bodies, intermediate filaments and thin filaments make the smooth muscle fiber shorten when sliding occurs between thick and thin filaments.

- ***Sarcotubular System***
- Sarcotubular system is in the form of network. ‘T’ tubules are absent and ‘L’ tubules are poorly developed



(a) Relaxed smooth muscle fiber (note that gap junctions connect adjacent fibers)



(b) Contracted smooth muscle fiber

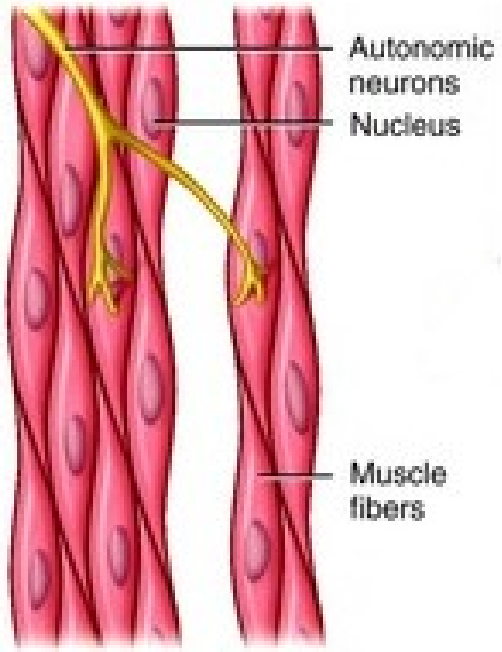
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TYPES OF SMOOTH MUSCLE FIBERS

- Single-unit or visceral smooth muscle fibers
- Multiunit smooth muscle fibers.

SINGLE-UNIT OR VISCERAL SMOOTH MUSCLE FIBERS

- The fibers with interconnecting gap junctions. The gap junctions allow rapid spread of action potential throughout the tissue.
- All the muscle fibers show synch contraction as a single unit.



(a) Visceral (single-unit) smooth muscle tissue

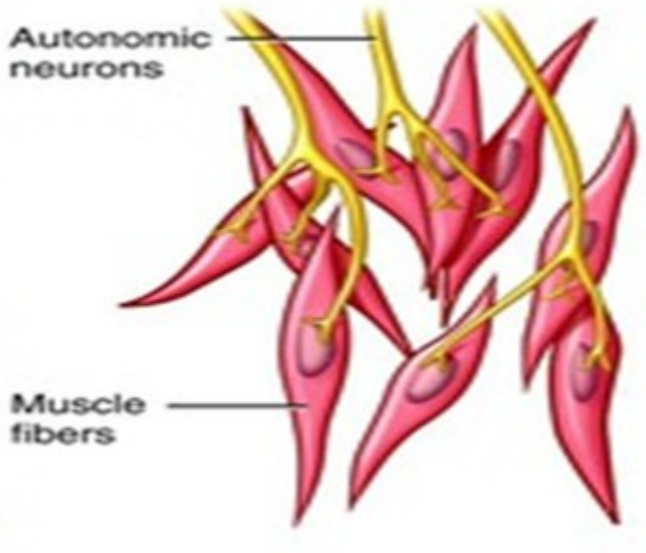
Distribution

In the walls of the organs such as

- Gastrointestinal organs,
- uterus,
- ureters,
- respiratory tract, etc.

MULTIUNIT SMOOTH MUSCLE FIBERS

- Multiunit smooth muscle fibers are the muscle fibers without **interconnecting gap junctions.**
- These smooth muscle fibers resemble the skeletal muscle fibers.



Distribution

- Ciliary muscles of the eye,
- Iris of the eye,
- Nictitating membrane (in cat),
- Arrector pili
- Smooth muscles of the blood vessels
- urinary bladder.

ELECTRICAL ACTIVITY

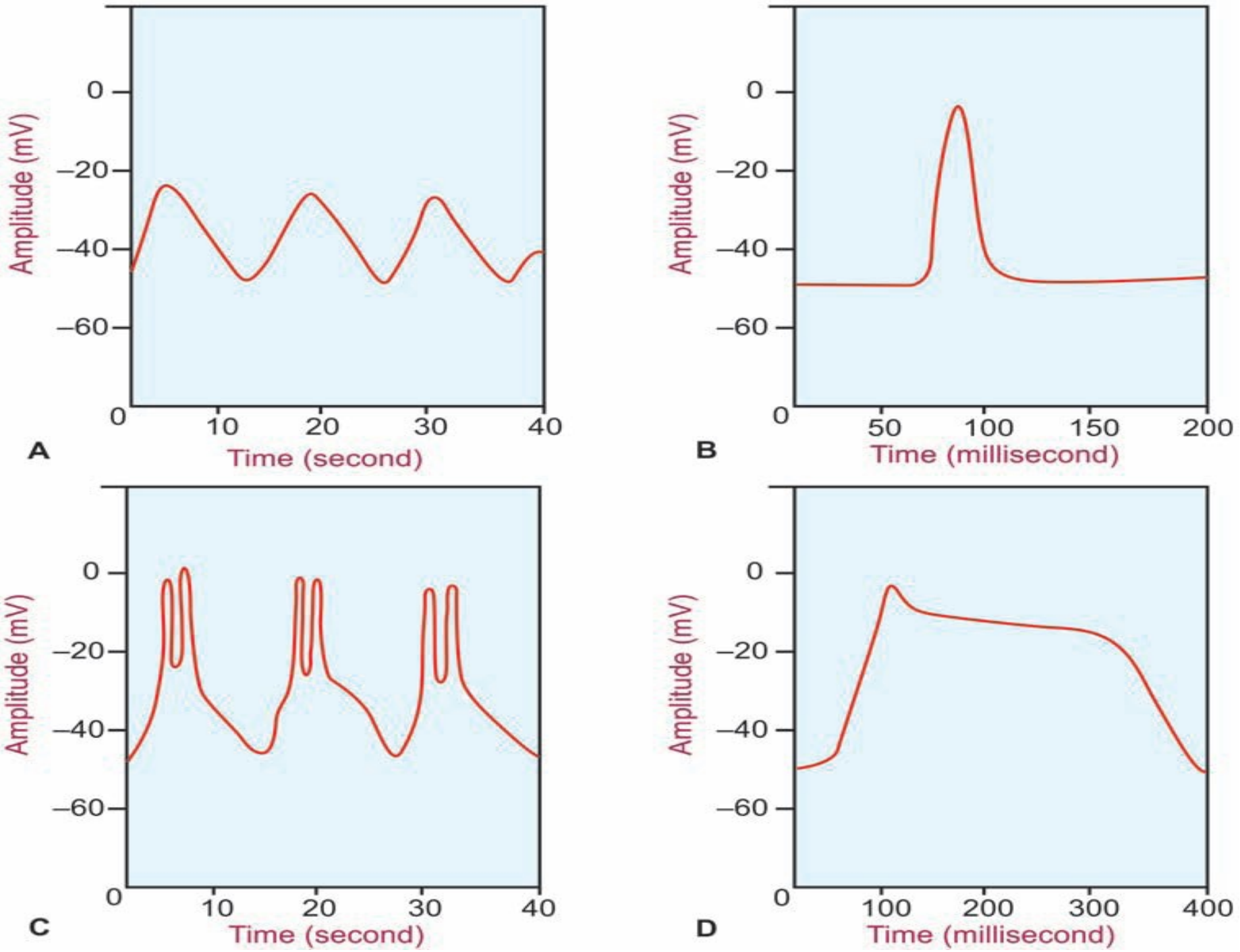
- ***Spike Potential***
- Similar to that of skeletal muscle.
- It is different from the spike potential in skeletal muscles in many ways.
- In smooth muscle, the average duration of spike potential varies between 30 and 50 milliseconds.
- Its amplitude is very low and it does not reach the isoelectric base.
- Sometimes, the spike potential rises above the isoelectric base (overshoot).
- The spike potential is due to nervous and other stimuli and it leads to contraction of the muscle.

Spike Potential Initiated by Slow-wave Rhythm

- Sometimes the slow-wave rhythm of resting membrane potential initiates the spike potentials, which lead to contraction of the muscle.
- The spike potentials appear rhythmically at a rate of about one or two spikes at the peak of each slow wave.
- This type of potentials appears mostly in smooth muscles, which are self-excitatory and contract themselves without any external stimuli.
- So, the spike potentials initiated by slow-wave rhythm are otherwise called **pacemaker waves.**
- The smooth muscles showing rhythmic contractions are present in some of the visceral organs such as intestine.

Action Potential with Plateau

- This type of action potential starts with rapid depolarization as in the case of skeletal muscle.
- Repolarization does not occur immediately.
- The muscle remains depolarized for long periods of about 100 to 1,000 milliseconds.
- Responsible for sustained contraction of smooth muscle fibers.
- After the long depolarized state, slow repolarization occurs


Amplitude (mV)
0
–20
–40
–60
0
10
20
30
40
Time (second)
A
Amplitude (mV)
0
–20
–40
–60
0
50
100
150
200
Time (millisecond)
B
Amplitude (mV)
0
–20
–40
–60
0
10
20
30
40
Time (second)
C
Amplitude (mV)
0
–20
–40
–60
0
100
200
300
400
Time (millisecond)
D

TONIC CONTRACTION OF SMOOTH MUSCLE WITHOUT ACTION POTENTIAL

- Smooth muscles of some visceral organs maintain a state of partial contraction called **tonus** or **tone.**
- It is due to the tonic contraction of the muscle that occurs without any action potential or any stimulus.
- Sometimes, the tonic contraction occurs due to the action of some hormones.

IONIC BASIS OF ACTION POTENTIAL

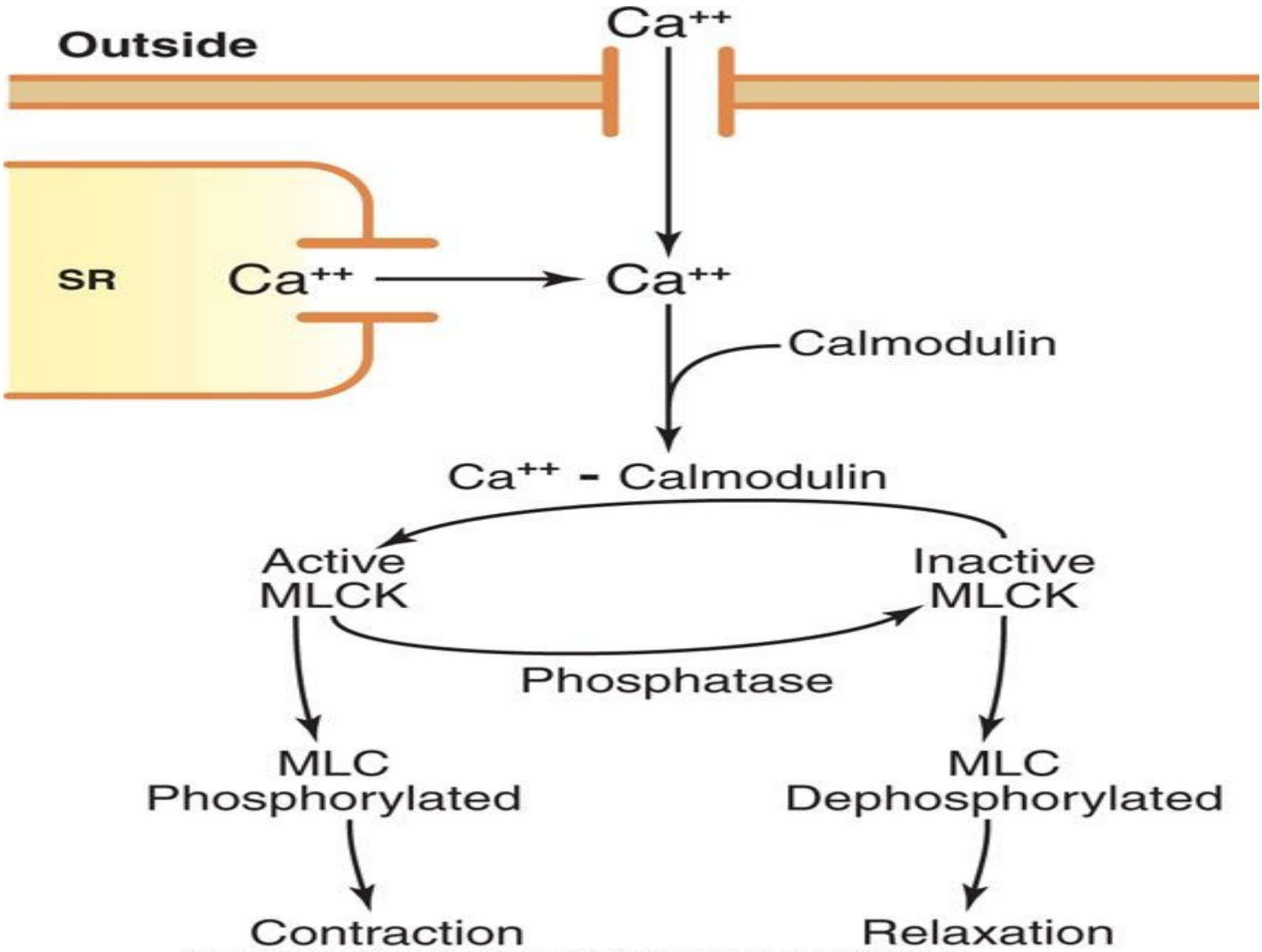
- The depolarization is due to entry of calcium ions rather than sodium ions.
- The calcium channels open and close slowly.
- It is responsible for the prolonged action potential with plateau in smooth muscles.
- The calcium ions play an important role during the contraction of the muscle.

MOLECULAR BASIS OF SMOOTH MUSCLE CONTRACTION

- Because of poor development of ‘L’ tubules (sarcoplasmic reticulum).
- The calcium ions, must be obtained from the extracellular fluid.
- It makes the process of excitation-contraction coupling slow.
- ***Calcium-calmodulin Complex***
- In smooth muscle, the myosin has to be phosphorylated for the activation of myosin ATPase.

Phosphorylation of myosin

- 1. Calcium, combines with a protein called calmodulin and forms calcium-calmodulin complex
- 2. It activates calmodulin-dependent myosin light chain kinase
- 3. This enzyme in turn causes phosphorylation of myosin followed by activation of myosin ATPase
- 4. Now, the sliding of actin filaments starts.
- Phosphorylated myosin gets attached to the actin molecule for longer period.
- It is called **latch-bridge mechanism** and it is responsible for the sustained contraction of the muscle with expenditure of little energy.
- Relaxation of the muscle occurs due to dissociation of calcium-calmodulin complex.



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Length-Tension Relationship – Plasticity

- **Plasticity is the adaptability of smooth muscle fibers to a wide range of lengths. If the smooth muscle fiber is stretched, it adapts to this new length and contracts when stimulated.**
- **The property of plasticity in smooth muscle fibers is especially important in digestive organs such as stomach, which undergo remarkable changes in volume.**
- **In spite of plasticity, smooth muscle fibers contract powerfully.**
- **Smooth muscle fibers also show sustained tetanic contractions.**

CONTROL OF SMOOTH MUSCLE

- Smooth muscle fibers are controlled by:
- 1. Nervous factors
- 2. Humoral factors.
- 🇮🇳 **NERVOUS FACTORS**
- Smooth muscles are supplied by both sympathetic and
- parasympathetic nerves, which antagonize (act opposite
- to) each other and control the activities of smooth muscles.
- However, these nerves are not responsible for the
- initiation of any activity in smooth muscle.
- The tonus of smooth muscles is also independent of nervous control.

Hormones and Neurotransmitters

- **Action depends upon the type of receptors present in membrane of smooth muscle fibers in particular area.**
- If excitatory receptors are present, contract the muscle by producing depolarization.
- If inhibitory receptors are present, relax the muscles by producing hyperpolarization.
- Hormones and neurotransmitters, which act on smooth muscles are:
- **1. Acetylcholine, 2. Antidiuretic hormone (ADH)**
- **3. Adrenaline, 4. Angiotensin II, III and IV, 5. Endothelin,**
- **6. Histamine, 7. Noradrenaline, 8. Oxytocin, 9. Serotonin.**

Other Humoral Factors

- Humoral factors which relax the smooth muscles:
- 1. Lack of oxygen
- 2. Excess of carbon dioxide
- 3. Increase in hydrogen ion concentration
- 4. Adenosine
- 5. Lactic acid
- 6. Excess of potassium ion
- 7. Decrease in calcium ion
- 8. Nitric oxide (NO), the **endothelium-derived relaxing**
- **factor** (EDRF).

THANK YOU